Utilizing Topic Modeling for Music Recommendation

Khoa Le

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Abstract

In the digital age, music streaming services face the challenge of providing personalized recommendations to millions of users while avoiding popularity bias. This thesis explores an alternative approach based on topic modeling of song lyrics. We compare three models—LDA, BERTopic, and Contextualized Topic Models (CTM)—to generate document-topic and artist-topic distributions, and evaluate their performance in terms of coherence, diversity, and recommendation quality. Our pipeline extracts lyrics via the Genius API, preprocesses text, trains topic models, aggregates song-level distributions into artist profiles, and uses cosine similarity with Maximal Marginal Relevance (MMR) for generating diverse recommendations. Experimental results demonstrate that CTM outperforms other models in both topic coherence and diversity, leading to more balanced and novel music suggestions. We conclude with discussion on integrating audio features and future improvements such as user studies and web interface deployment.

1 Introduction

Recommendation systems are integral to modern digital services, reducing information overload by filtering content based on user preferences [Adomavicius and Tuzhilin, 2005]. In music streaming platforms like Spotify and Apple Music, personalized suggestions help users discover new tracks, increasing engagement and retention[Celma, 2010]. However, popular algorithms often rely on collaborative filtering and audio metadata, which can amplify popularity bias, leading to homogeneous recommendations dominated by mainstream artists [Abdollahpouri et al., 2019].

Lyrics-based recommendation offers a complementary signal, focusing on thematic content rather than popularity. Unlike audio features or metadata, topic modeling of lyrics captures semantic patterns that reflect an artist's style and themes. This approach reduces dependence on external metadata and mitigates the tendency to over-favor popular artists. Our work investigates the efficacy of topic modeling for music recommendation, comparing classical and neural methods to deliver diverse, coherent suggestions.

2 Background

2.1 Recommendation Systems

Recommendation systems can be categorized into content-based, collaborative filtering, and hybrid approaches [Adomavicius and Tuzhilin, 2005]. Content-based methods analyze item attributes, while collaborative filtering leverages user-item interaction matrices. Hybrid models combine both to balance novelty and accuracy.

2.2 Music Recommendation

In music services, collaborative filtering is widely used due to abundant listening data, but suffers from cold-start and popularity bias issues [Celma, 2010, Abdollahpouri et al., 2019]. Content-based systems using audio features (e.g., timbre, rhythm, pitch) and metadata (e.g., genre, artist popularity) offer alternative signals but still favor common patterns found in popular tracks.

2.3 Popularity Bias

Popularity bias describes the phenomenon where popular items become more promoted, reducing exposure for niche content [Chen et al., 2020]. In automatic playlists on Spotify, users often get looped into a narrow set of top-chart songs, limiting music discovery.

2.4 Lyrics-based Recommendation

Lyrics capture thematic nuances of songs, from emotions to storytelling. Topic modeling can uncover latent themes across a corpus of lyrics, enabling recommendations based on semantic similarity rather than play counts or collaborative signals.

3 Topic Modeling

3.1 Latent Dirichlet Allocation (LDA)

LDA [Blei et al., 2003] is a generative probabilistic model that represents each document as a mixture of topics, with each topic being a distribution over words. It employs Dirichlet priors for document-topic and topic-word distributions, allowing unsupervised discovery of thematic structures.

3.2 BERTopic

BERTopic [Grootendorst, 2022] integrates transformer embeddings with clustering algorithms (e.g., HDBSCAN) and class-based TF-IDF to generate interpretable topics. It assumes each document belongs to a single cluster, limiting its ability to model multi-topic documents.

3.3 Contextualized Topic Models (CTM)

CTM [Charnois et al., 2021] extends classical topic models by incorporating contextual embeddings into the inference process. Using variational autoencoders, CTM captures both global topical distributions and local semantic contexts, improving coherence and diversity. ProdLDA [Srivastava and Sutton, 2017] serves as a precursor by applying neural variational inference to topic modeling.

4 Methodology

4.1 Dataset Construction

We assembled a corpus of 600 US-UK artists by querying ChatGPT and verified via the Genius API. For each artist, we extracted the top 10 songs' lyrics, resulting in 6,000 documents.

4.2 Lyrics Extraction

Using the Genius API, we fetched raw lyric texts. Non-English songs were filtered out during preprocessing. Metadata such as song title and artist name were retained for aggregation.

4.3 Preprocessing

Lyrics were cleaned by removing punctuation, lowercasing, and tokenizing. Stop words and infrequent tokens (`min_df`=5) were removed. Lemmatization was applied using part-of-speech tagging to improve normalization.

4.4 Model Training

We trained LDA (`gensim`), BERTopic, and CTM (via `OCTIS`) with $K = 10$ topics. Hyperparameters were tuned on a held-out validation set.

4.5 Artist-level Aggregation

Document-topic distributions (θ_d) were averaged per artist to form artist-topic profiles ($\Theta_a = \frac{1}{|D_a|} \sum_{d \in D_a} \theta_d$), reflecting an artist's thematic signature.

4.6 Recommendation Generation

Given a query artist or song, we computed cosine similarity between profiles. For diversified recommendations, we applied Maximal Marginal Relevance (MMR) [Carbonell and Goldstein, 1998] to re-rank candidates, balancing relevance and novelty.

5 Evaluation

5.1 Topic Coherence and Diversity

We measured coherence (UMass) and topic diversity [Dieng et al., 2020]. CTM achieved an average coherence of 0.45 and diversity of 0.72, outperforming LDA (0.31, 0.58) and BERTopic (0.22, 0.49).

5.2 Recommendation Quality

Quantitative evaluation used intralist similarity reduction via MMR and qualitative inspection of top recommendations. CTM-based suggestions showed thematic relevance and diversity, avoiding repetitive mainstream artists.

6 Results

Across 100 random artist queries, CTM recommendations had 30% lower similarity to top-10 popular lists compared to LDA, indicating reduced popularity bias. BERTopic often collapsed into 2 topics (English vs non-English) without fine-tuning, limiting its utility.

7 Discussion

Our findings demonstrate that contextualized topic models effectively capture lyrical themes, producing coherent and diverse artist recommendations. Limitations include lack of user feedback, reliance on lyric availability, and exclusion of audio features.

8 Conclusion and Future Work

We presented a lyrics-based music recommendation pipeline leveraging topic modeling. Future directions include user studies for subjective evaluation, web interface deployment, and integration of audio embeddings for hybrid recommendations.

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